

03

Digital Electronics & VLSI
Codec Technologies India
Internship 2026

Smart Digital Lock in VHDL-2008

Keypad scanner, PIN FSM with
lockout and change-PIN,
7-segment status, iCE40 P&R;

AT A GLANCE

38 / 38 pass

GHDL testbench checks

88

Key presses simulated

439 / 7680

iCE40 logic cells (nextpnr)

97.21 MHz

iCE40 Fmax (nextpnr)

10.80 ns

icetime longest path

180 / 182

Xilinx LUT / FF (yosys)

30 s after 3 wrong

Lockout

8

FSM states

VERDICT

SIM PASS

P&R OK

TIMING MET

GHDL PLUGIN USED

DECLARED SUBSTITUTION

Eswar Mahalingam

VLSI / Digital Design Intern
Ghaziabad, UP, India
+91-9360548243
eswarmba05313@gmail.com
linkedin.com/in/eswar-mahalingam

1. Objective and brief mapping

The brief asked for a VHDL-2008 digital lock with a 4x4 matrix keypad scanner (row drive, column sense, debounce), a 4-digit PIN entry FSM with the PIN in an internal register and a change-PIN mode, lockout with alarm and a 30 s timeout after three wrong attempts, a master reset, a dual 7-segment multiplexer showing entered digits or status codes (OP, Er, Lo), a self-checking GHDL testbench with assert/severity failure covering correct PIN, wrong PIN x3, master reset and PIN change, waveform PNGs, synthesis through the yosys ghdl plugin if available (else ghdl --synth), and Vivado tcl/xdc declared not run.

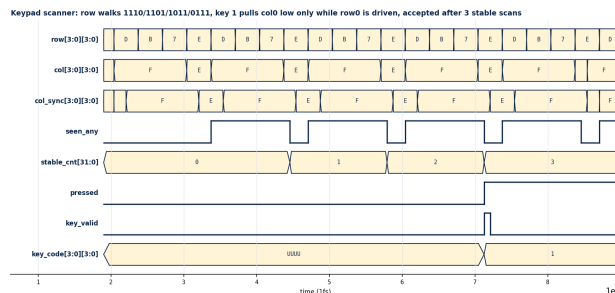
All of it is implemented. The ghdl plugin was tested first and works, so the VHDL was synthesised directly by yosys -m ghdl. Every number here was produced in the build container by GHDL 7.0.0-dev, yosys 0.68, nextpnr-ice40 and icetime.

2. Architecture

```
col[3:0] -> 2-FF sync -> keypad_scanner (row walk 1110/1101/1011/0111, SCAN_DIV clk per row,
DEB_SCANS stable scans, release required) -> key_code[3:0] + key_valid pulse
12 MHz -> tick (1 Hz) master_rst_n -> sync + 10 ms debounce -> master_rst

pin_fsm: IDLE -digit-> ENTER -4 digits-> match? OPEN(OP, unlock) : ERROR(Er, attempts+1)
ERROR & attempts=3 -> LOCKOUT(Lo, alarm, 30 s) -> IDLE OPEN -A-> NEW1 -> NEW2 -> CH
sym_hi/sym_lo (5-bit symbol codes) -> seg7_mux (anode alternates every REFRESH_DIV) -> s
```

Block	Key design points
keypad_scanner	One-cold row drive; columns synchronised through two flops; each row held SCAN_DIV clocks before sampling; a key is accepted once after DEB_SCANS identical full scans and only after a release, so contact bounce and held keys cannot repeat. Legend 1 2 3 A / 4 5 6 B / 7 8 9 C / * 0 # D encoded 0-9, A-D, E(*), F(#).
pin_fsm	Eight states; pin_reg (4 x BCD) initialised to DEFAULT_PIN; entry shift register with digit count; * clears, 4th digit or # submits; attempts counter (2 bits) cleared on a correct PIN, on lockout expiry and on master reset; one shared seconds timer reloaded on each timed state.
seg7_mux + lock_pkg	Symbol codes 0-9 plus dash, O, P, E, r, L, o, C, h, n; the code-to-segment function lives in the package so the testbench decodes the physical segment bus with the same table.
lock_top	Reset button synchronised; master reset synchronised and debounced (MR_DEB); 1 Hz tick divider; state_led and attempts_led exported as debug LEDs and used by the testbench.



Scanner detail for key 1: the row pattern walks 1110/1101/1011/0111 every 4 clocks; col0 drops only while row0 is low; stable_cnt counts identical scans and key_valid fires once when it reaches 3, pressed then blocks repeats until release.

3. Verification

tb_lock_top.vhd contains a behavioural key matrix that pulls the pressed key's column low only while its row is driven, so the scanner is tested through the real row/column protocol. Every expectation is an assert with severity failure and the run exits non-zero on the first miss. Because VHDL-2008 external names crash at run time in this GHDL build, the bench observes only the top-level ports: state_led, attempts_led, unlock, alarm and the multiplexed segment bus, which it decodes digit by digit. Time bases are scaled (1 s = 100 clocks, one row = 4 clocks, debounce = 3 scans) but the lockout keeps its real value of 30 s. Result: 88 key presses, 38 checks, PASS.

Scenario	Checks	Result
T1 correct PIN 1234	display -1, 12, 23 while typing; OPEN with unlock; OP shown; auto-relock after T_OPEN	pass
T2 bounce, *, D	1-scan bounce not accepted; * clears entry to --; 1234 accepted; D relocks immediately	pass
T3 wrong PIN x3	Er with attempts 1, 2, 3; LOCKOUT with alarm and Lo; correct PIN ignored in lockout; expiry after 30 ticks clears attempts	pass
T4 master reset	second lockout cleared by master reset: IDLE, alarm 0, attempts 0	pass
T5 PIN change	A from OPEN, 5678 twice -> Ch; 1234 rejected; 5678 opens and clears attempts	pass
T6 mismatch / restore	1111 vs 2222 -> Er without counting an attempt, PIN unchanged; master reset restores 1234	pass

03

Digital Electronics & VLSI
Codec Technologies India
Internship 2026

Smart Digital Lock in VHDL-2008

Keypad scanner, PIN FSM with
lockout and change-PIN,
7-segment status, iCE40 P&R;

AT A GLANCE

38 / 38 pass

GHDL testbench checks

88

Key presses simulated

439 / 7680

iCE40 logic cells (nextpnr)

97.21 MHz

iCE40 Fmax (nextpnr)

10.80 ns

icetime longest path

180 / 182

Xilinx LUT / FF (yosys)

30 s after 3 wrong

Lockout

8

FSM states

VERDICT

SIM PASS

P&R OK

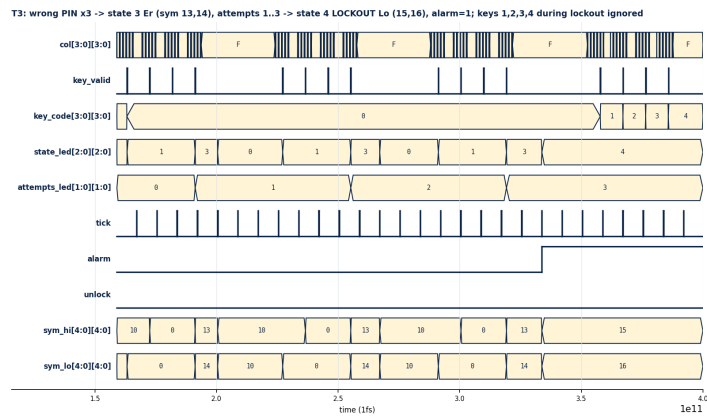
TIMING MET

GHDL PLUGIN USED

DECLARED SUBSTITUTION

Eswar Mahalingam

VLSI / Digital Design Intern
Ghaziabad, UP, India
+91-9360548243
eswarmba05313@gmail.com
linkedin.com/in/eswar-mahalingam

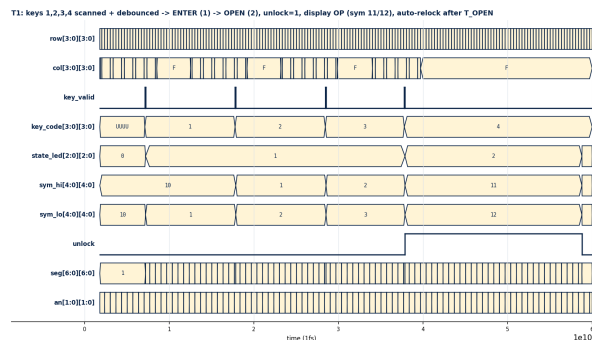


T3 in GHDL: three 0000 entries drive state 1 -> 3 (Er) with attempts_led 1, 2, 3; the third error drops into state 4 (LOCKOUT), alarm rises and the display shows Lo (symbols 15/16); the 1, 2, 3, 4 presses during lockout produce key_valid pulses but no state change.

4. Synthesis and implementation results

Roughly two thirds of the flip-flops are time-base counters (24-bit tick divider, 17-bit master-reset debounce, 14-bit row divider, 13-bit refresh divider); the lock logic itself is about 60 flops. The critical path is the 16-bit compare of the confirmation entry against new_pin feeding the pin_reg load enable, which is why icetime ends at u_fsm.new_pin. At 12 MHz the slack is more than 70 ns, so there is room to run the whole design from a 100 MHz Basys3 clock without changes other than the generics.

Target / tool	Resources	Timing
iCE40-HX8K, yosys -m ghdl synth_ice40	361 SB_LUT4, 182 FF (12 DFF, 39 DFFE, 69 DFFESR, 1 DFFESS, 61 DFFSR), 67 SB_CARRY	-
iCE40-HX8K CT256, nextpnr-ice40, real .pcf	439 / 7680 logic cells (5 %), 27 / 206 IO	Fmax 97.21 MHz, 12 MHz PASS
iCE40-HX8K, icetime -mtr	8 logic levels, endpoint u_fsm.new_pin DFFE	10.80 ns (92.63 MHz)
Xilinx xc7, yosys synth_xilinx	180 LUT (50 LUT2, 10 LUT3, 34 LUT4, 54 LUT5, 32 LUT6), 182 FF, 23 CARRY4	not placed (no Vivado)



T1: four accepted keys (key_code 1,2,3,4) take state_led from 0 to 1 (ENTER) to 2 (OPEN); unlock rises, sym_hi/sym_lo switch to 11/12 (OP) and the multiplexed seg/an bus alternates between the two digits; auto-relock returns to IDLE after T_OPEN.

5. Declared substitutions and how to run on the target tools

- Vivado / Basys3: not installed here. vivado -mode batch -source vivado/create_project.tcl creates the project with the VHDL-2008 sources, vivado/basys3.xdc (W5 100 MHz clock, keypad on Pmod JA with PULLUP, on-board 7-segment and LEDs) and the 100 MHz generics, then writes utilization and timing reports. The Xilinx LUT/FF numbers above are from yosys synth_xilinx, not Vivado.
- No physical keypad or board: the keypad is a behavioural matrix model; iCE40 figures come from nextpnr/icetime on the HX8K device database; icepak produced the bitstream during make synth to prove the flow completes.
- The fallback ghdl --synth -> yosys path described in the brief was not needed because the ghdl plugin was present; it remains a one-line change in synth/synth_ice40.ys.

6. Learnings and next steps

Two things cost me time and are worth writing down. First, VHDL is case-insensitive, so SYM_O and SYM_o collided in the package; lower-case letter symbols now carry an explicit LO_ prefix. Second, hierarchical external names are convenient but not portable: when they failed at run time I moved to a port-only testbench, which is also closer to how the board will be debugged with a logic analyser. Next steps: store the PIN in the iCE40 SPI flash or an I2C EEPROM so it survives power loss, add a tamper input that forces lockout, and blank the entered digits after a configurable delay for shoulder-surfing resistance.